

Proposal and Evaluation of A Method for Automatically Classifying Images of Agricultural Work and Animals Acquired with Motion Sensor Cameras

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Abstract—We have developed a field monitoring system (<http://kansatu.net>) using network-connected cameras and sensors installed in apple trees to support the hands-on agriculture curriculum of an elementary school. The system enables to collect and store a variety of images taken by network-connected cameras equipped with infrared motion sensors. Agricultural works and animals are photographed in these images. The system has been in place seven years since 2011, during which time, there have been years in which more than approximately 50,000 images were collected. However, these data have not been used effectively to learn agricultural works. On the other side, in recent years, damage to agricultural crops by wild animals has been increasing. From these backgrounds, our ultimate goal is to utilize the images captured by motion sensor-equipped network cameras to develop countermeasures to prevent wild animals from damaging agricultural crops and to educate and cultivate future farmers. In this study, we proposed a method that employs image analysis technology and the date and time of image capture to automatically classify images acquired by the motion sensor network cameras by type of agricultural work and animal. We also developed a method for automatically classifying the type of agricultural work. We evaluated the accuracy of the developed method by comparing the results of automated classification with the results of manual classification. The recall of the proposed method exceeded 90% for all three types of agricultural work tested, which was equal to or greater than the classification accuracy achieved with manual classification.

Keywords—field monitoring system; motion sensor camera; image processing; agricultural information

I. INTRODUCTION

In recent years, considerable advances have been made in the research and development of commercial field monitoring systems. However, the use of this technology has been limited to environmental monitoring in greenhouse production and quality control of agricultural products. Moving into a different area of application, we have developed a field monitoring system (<http://kansatu.net>) using cameras and sensors installed in outdoor apple trees to support the hands-on agriculture curriculum of an elementary school. The system, which enables the young students to observe the trees and fruit from inside the classroom, collects and stores a variety of environmental data, including

temperature, humidity, and quantity of solar radiation, as well as fixed-point images of agricultural crops, and images taken by network-connected cameras equipped with infrared motion sensors. The system has been in place seven years since 2011, during which time, there have been years in which more than approximately 50,000 images were collected. In addition, a vast amount of environmental data, recorded in 1- or 2-min intervals, has been accumulated. However, the reality is that these data have not been used effectively [1].

Meanwhile, Japanese agriculture is faced with an aging farming population and is struggling to secure new farmers to take over farm operations. The number of individuals engaged in agriculture is declining from year to year. According to a survey conducted in 2015, 63.5% of the farming population was aged 65 or older, with mean age being 66.4 years[2]. In addition, in recent years, damage to agricultural crops by wild animals has been increasing. According to a report from the Ministry of Agriculture, Forestry and Fisheries (MAFF), the damage to agricultural crops caused by birds and wild animals in 2012 was substantial, amounting to an estimated 23 billion JPY. Damage caused by birds and wild animals is also a problem in Iwate Prefecture, which is the field site of this study.

Against this background, our ultimate goal is to utilize the images captured by motion sensor-equipped network cameras (hereinafter called “motion-triggered images”) to develop countermeasures to prevent wild animals from damaging agricultural crops and to educate and cultivate future farmers. As a first step towards this goal, in this study, we propose a method that employs image analysis technology and the date and time of image capture to automatically classify images acquired by the motion sensor network cameras installed in the field by type of agricultural work and animal. If the captured images can be automatically classified by type of agricultural work or animal, they can then be used as training materials and to develop animal deterrence methods.

II. CONCEPT OF THE SYSTEM USING MOTION SENSOR IMAGES

A. Previous Research

In a previous study, two network cameras (BB-SW174W, Panasonic) and a mobile Wi-Fi router (Wi-Fi Station N-01H,

NTT DOCOMO) were installed in an apple orchard in Shiwa, Iwate Prefecture and used to monitor apple fruit and whole apple trees. Each camera captured one image every hour from 5 am to 6 pm. In FY2012, additional network cameras equipped with motion sensors (MS-12FA, Takeuchi Engineering) were installed to automatically capture images of field work. Upon detecting motion, the motion sensor cameras capture one image per second for the next 100 seconds. If motion is detected during the 100-s period, the cameras are set to capture images for another 100 s, thereby storing a detailed record of agricultural work. A field server (FS-V, eLAB experience) was also installed, and collection of environmental data was started in FY2012. In FY2014, the server was removed, and temperature, solar radiation, humidity, and vapor pressure deficit data were stored using a Midori box [green box] (Seraku Co., Ltd.) [3]. Trail cameras were installed in 2017 to capture images of wild animals, both during the day and at night.

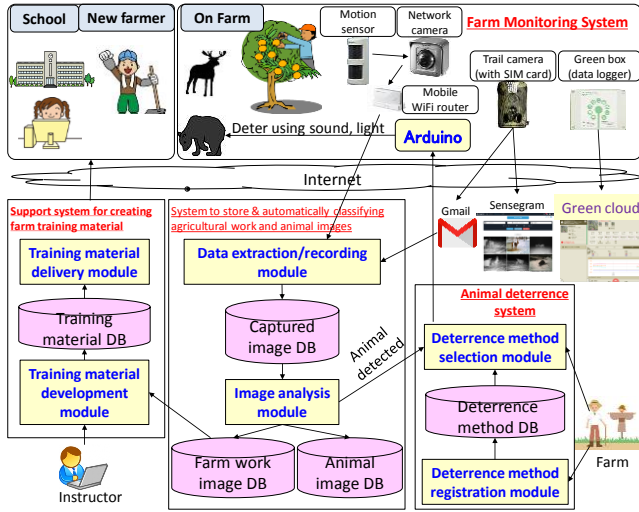


Figure 1. " Example of a field monitoring system application.

B. Ultimate Goal

A conceptual diagram of the system for using motion-triggered images recorded by the previously developed field monitoring system is shown in Fig. 1. In this study, we conceive of a support system for using motion-triggered images to develop training materials for various types of agricultural work and a support system for using motion-triggered images to select and implement appropriate animal deterrence methods based on the animal type and past history.

III. OVERVIEW OF IMAGE ANALYSIS AND AUTOMATIC CLASSIFICATION

A. Image Analysis

The numbers of images captured in 2011 to 2017 are shown in Table 1. As the cameras in this system are installed outdoors, there were periods when the cameras could not capture images due to overheating. For this reason, the number of images captured differed from year to year.

TABLE I. NUMBER OF IMAGES CAPTURED ANNUALLY

Fiscal Year	Fruit	Trees	Motion-triggered images	Total
FY2011	3,762	4,213	0	9,986
FY2012	5,063	4,779	11,305	23,159
FY2013	4,371	4,395	22,321	33,100
FY2014	4,067	4,054	30,524	40,659
FY2015	4,494	4,499	39,396	50,404
FY2016	2,964	3,136	29,951	38,067
FY2017	0	2,592	3,813	8,422

TABLE II. CATEGORIES OF AGRICULTURAL WORK CAPTURED IN MOTION-TRIGGERED IMAGES

Work category	Work content
Pruning	Adjustment of branch arrangement and tree vigor
Fertilizing	Promotion of tree development by applying fertilizer
Flower thinning	Adjustment of flower quantity and quality (before flowering)
Fruit thinning	Adjustment flower bud quantity and quality (after flowering)
Insect pest control	Removal of insect pests by applying insecticide
Weed management	Removal of weeds that hinder tree growth
Leaf thinning / fruit rotating	Leaf removal and rotation of fruit to allow sunlight to reach all sides of fruit (which promotes even coloring).
Laying of reflective sheets	Light reflected by the reflective sheet colors the bottoms of fruit.
Fruit harvest and sorting	After harvest, fruit are sorted into five levels based on quality
Other work	Erroneous detection (no people present in image), people passing by, camera checks, excessive backlighting, blizzards, insufficient light to categorize work type, work type unknown, image defects, non-agricultural work, etc.

To investigate methods for automatically classifying motion-triggered images of agricultural work, we first analyzed motion-triggered images of agricultural work. A total of 22,231 motion-triggered images captured between January 2013 and December 2013 were analyzed using the three procedures below.

- (1) Determination of agricultural work content based on comparison of image capture time and agricultural work records.
- (2) Determination of agricultural work content based on crop changes.
- (3) Determination of agricultural work content based on tools, implements, etc. used.

Using the above methods, we were able to classify the motion-triggered images into 10 work categories (Table 2). Excluding "other," the nine categories comprised basic work associated with apple cultivation. As such, we expected that the images would be useful for teaching how to cultivate apples.

B. Overview of Automated Image Analysis

In this study, we are developing a system that automatically determines whether a motion-triggered image is of agricultural work or an animal. If the image is of agricultural work, the system adds metadata regarding the specific type of work (Table 2). If there is a chance that, when a user searches for agricultural work images based on the added metadata, some number of relevant images will not be included in the search results, it becomes necessary to search through all captured images. In that case, the amount of effort required is the same as when the metadata system is not used. To ensure that that is not the case, we must make sure that not a single relevant image is missed (recall of 100%) when the user searches agricultural work images. It is generally accepted that there is a tradeoff between recall and precision. For this reason, if the recall is 100%, the precision is expected to be low. Accordingly, in this study, we investigate methods to minimize the number of irrelevant images (i.e., improve precision) while ensuring that all relevant images are found (recall of 100%).

IV. PROPOSAL AND DEVELOPMENT OF METHOD FOR CLASSIFYING AGRICULTURAL WORK AND ANIMAL IMAGES

A. Proposal of Method for Classifying Agricultural Work and Animal Images

In this study, we propose the following six methods for classifying motion-triggered images by type of agricultural work and animal.

- (1) Determination of presence of agricultural worker or animal
- (2) Determination based on time of year (January to March)
- (3) Determination based on change in crops
- (4) Determination based on equipment/implement used
- (5) Determination based on time of year (April to December)
- (6) Determination of animal type

The procedure for automatically classifying motion-triggered images is shown in Fig. 2. Metadata is added to motion-triggered images based on the results of this classification. This will enable identification of a single type of agricultural work or a narrowing of choices to a smaller number of possible work types even in cases where manual determination by visual inspection is difficult. For example, if metadata is not added in Steps (1) to (4) of the procedure shown in Fig. 2 (i.e., if (1) a worker is determined to be present, (2) the image is captured between April and December, (3) no crop change is detected, and (4) no equipment or tools are identified), determination will be made based on (5) time of year (April to December). For example, if the image capture date is September 25, based on the annual work schedule shown in Table 3, the metadata added will be “leaf thinning/fruit rotation,” “laying of reflective sheets,” and “fruit harvest and sorting.” Determination methods (1) to (5) are explained in greater detail below. Determination method (6) is currently being investigated.

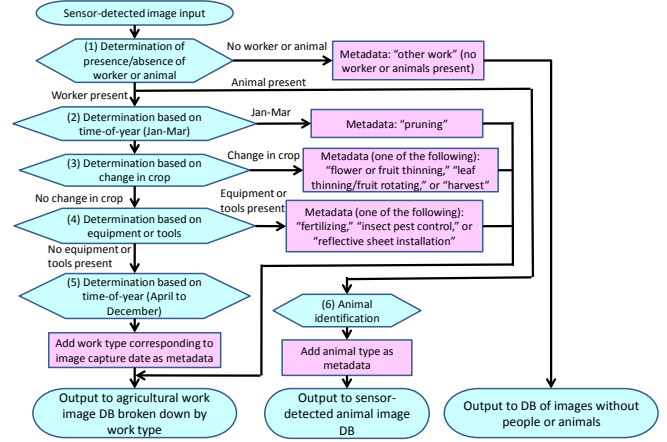


Figure 2. Procedure for automatically classifying motion-triggered images.

TABLE III. ANNUAL SCHEDULE FOR MAIN AGRICULTURAL WORK

Work category	Work period
Pruning	January to March
Fertilizing	April
Flower thinning	May to June
Fruit thinning	May to June
Insect pest control	April to August
Weed management	May to August
Leaf thinning / fruit rotation	August to November
Laying of reflective sheets	August to November
Fruit harvest and sorting	September to November
Other work	Year-round

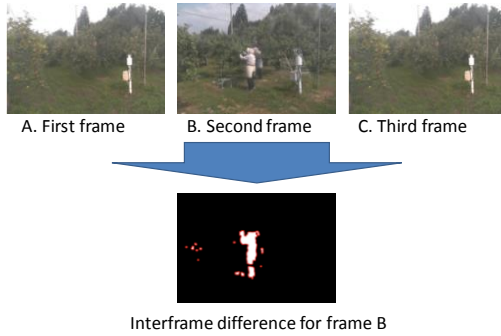
B. Determination of Agricultural Worker or Animal Presence/Absence

Among the motion-triggered images, there are images in which no people or animals are present or, even if people are present, they are not performing agricultural work. In such cases, the metadata added is “other.”

1) Determination based on the interframe difference method

Motion-triggered images captured on the same day are arranged in chronological order, and interframe differences are evaluated based on the method below. Consecutive images in which no interframe differences are detected are determined to not contain people or animals.

- (1) For n images captured on the same day, sets of three consecutive images are acquired starting with image 1 and ending with image $n-2$.
- (2) Differential images are created for the first and second images and for the second and third images of each triplet.
- (3) Logical products of the two differential images are calculated.



$$AB \text{ differential} \cap BC \text{ differential} = \text{interframe difference for frame B}$$

Figure 3. " Example of the interframe differential method to detect presence of a person.

- (4) If the differential area does not exceed the threshold of 0.5 for any of the $n-2$ image triplets, the metadata added is "other."

Substantial interframe differences are used to determine presence of environmental change outdoors. The images are then subjected to labeling process (labeling classes used by the Imura Laboratory) [4], and a determination is made whether images exist for which the maximum differential area exceeds the threshold value. In this case, the threshold is set to 0.5, based on the value where presence or absence of people or animals could be determined even when dust or water droplets were present on the camera lens. Images captured at night are classified as animal images, and images capture during the day are classified as agricultural work images. An example of the interframe differential method used to detect the presence of a person is shown in Fig. 3.

2) Determination based on the motion sensor trigger count

Among the images identified by the interframe differential method as containing agricultural workers, there are cases in which workers are simply passing through or are checking the camera but are not performing any agricultural work. Workers who are in the area for purposes other than agricultural work tend to stay only a short period of time. Accordingly, a determination can be made that a person is not performing agricultural work based on the number of times the motion sensor is triggered. In such cases, the metadata added is "other."

The results of an analysis to determine the relationship between presence/absence of agricultural work and motion sensor trigger count in motion-triggered images captured in FY2013 are shown in Table 4. On days when some type of agricultural work is being performed, the trigger count is three or higher. Accordingly, in cases where the trigger count in a given day is two or less, a determination can be made that agricultural work is not being performed, and the metadata added is "other."

3) Determination based on time of year (January to March)

Most apple-related agricultural work—which includes pruning, flower thinning, leaf thinning and fruit rotation, harvest, and laying of reflective sheets—is performed at specific times of the year. Table 3 shows the time of year

that each type of agricultural work is performed. The table was created based on interviews with apple growers in Shiwa, Iwate Prefecture, which is the study site. From the table, it is evident that no work other than pruning is performed from January to March. Accordingly, "pruning" is added as metadata to motion-triggered images captured from January to March.

TABLE IV. RELATIONSHIP BETWEEN PRESENCE/ABSENCE OF AGRICULTURAL WORK AND MOTION

	No. of days detected	3 or more per day	2 or less per day
Agricultural work	122	122	0
No agricultural work	64	4	60
Total	186	126	60

4) Determination based on crop changes

Regarding work in which apples are directly handled—this includes harvest, flower thinning and fruit thinning, and leaf thinning/fruit rotation, the type of work is determined by evaluating changes in RGB values of the captured images. For example, the apple-colored area (red, in this study) in time-lapse images decreases dramatically after harvest since are no longer present. Similarly, the flower-colored are (white, in this study) decreases dramatically after flower thinning. For this reason, the day on which the greatest change (decrease) in red area is observed is determined to be the harvest day, and the day on which the greatest change (decrease) in white area is observed is determined to be the flower thinning day.

5) Determination based on equipment and tools

Motion-triggered images captured during agricultural work also show the equipment and tools used. Thus, it is possible to determine the type of work being performed by identifying the equipment or tools present. Determination based on equipment and tools utilizes the template matching algorithm of OpenCV (ver.2.4.1) to match captured images with pre-trained images. We selected OpenCV because it was found that similar images were captured for a given type of work, owing to the fact that the images are taken from a fixed position and the fact that machine-based work is generally performed in the same direction along the same path.

In this determination method, the equipment to be identified includes an agricultural chemical sprayer and a riding lawnmower. Images identified as containing these machines were assigned "Insect pest control" and "Weed management," respectively, as metadata. In addition, as it was found that the angle from which these machines were captured differed depending on the camera direction seven template images were prepared for each piece of equipment. Examples of template images for the agricultural chemical sprayer are shown in Fig. 4. Images of the same equipment captured from various angles in different years were used as template images to improve detection accuracy. Furthermore, because the motion sensor cameras are installed outdoors, the captured images are not necessarily clear since there is a possibility of rain or dust being on the lens. Accordingly, the

template matching threshold was adjusted and, in the end, set to 0.6, as this value was found to provide the highest matching accuracy. An example of an image identified as containing the agricultural chemical sprayer is shown in Fig. 5.



Figure 4. " Four representative template images for the chemical sprayer.



Figure 5. " Example of an image of the chemical sprayer identified by template matching.

6) Determination base on time of year (April to December)

If work type cannot be classified based on the methods above, work that was carried out on the image capture date is added as metadata. That said, from April to December, multiple types of work are performed around the same time. Accordingly, all work types that may have been performed on the image capture date are added as metadata, with the goal being to not lower the recall of searches for any given work type.

C. Development of Automated Agricultural Work Classification and Addition of Metadata

We developed a system for automatically classifying agricultural work using the above-listed methods and added metadata to all motion-triggered images that were captured. This metadata enables users to search for images captured

during agricultural work from among all motion-triggered images. The image analysis methods developed include template matching using OpenCV, frame differential analysis using labeling.h [4], and RGB analysis using C/C++.

V. EXPERIMENT

A. Experiment Overview

We performed an experiment to evaluate the agricultural work classification accuracy of the proposed methods ((1) to (5) in Fig. 1). In the experiment, we compared the results of manual classification and automated classification using the proposed method. The target images were images showing three representative types of agricultural work, namely weed management, insect pest control, and pruning. The test image pool consisted of 22,321 motion-triggered images captured in FY2013 that were identified as containing agricultural work of the types listed in Table 2.

The procedure for manual classification was as follows: First, all motion-triggered images were separated into folders on the PC desktop based on image capture date. Classification was performed by 11 students in the Faculty of Software and Information Science and the Graduate School of Software and Information Science of Iwate Prefectural University (3 second-year, 5 third-year, and 2 fourth-year undergraduate students and 1 graduate student). The classifiers first identified images showing pruning, followed by all images showing insect pest control, and finally all images showing weed management. The students were asked to check all images on the desktop and then to determine the type of work carried out each day. They were then asked to copy all images captured on a given day identified as showing a certain type of agricultural work to a different folder on the desktop labeled with the work type. The amount of time required to perform this work was calculated. In addition, the recall, precision, and F-value (harmonic mean of the recall and precision) were calculated for manual classification using the equations below.

$$\text{Precision factor} = \frac{\text{true positive}}{\text{total positive}}$$

$$\text{Recall factor} = \frac{\text{true positive}}{\text{actual true positive}}$$

$$\text{F-value} = \frac{2 * \text{recall} * \text{precision}}{\text{recall} + \text{precision}}$$

Next, from among the pool of motion-triggered images to which metadata was added by the proposed automatic classification method, we searched for images of the three work types that were classified manually. The recall, precision, and F-value were calculated for the automated classification method and compared with those for manual classification.

B. Experiment Results and Discussion

Results of the experiment are presented in Tables 5 to 7. The proposed method assumes that images captured over the course of one day will be analyzed and assigned metadata at

night. Although this analysis takes several minutes to perform, metadata will have already been added to the images by the time the user uses the system, so the time required for the proposed method (Tables 5 to 7) is considered 0. Tables 5 to 7 indicate that the time required for manual classification varied by classifier and by type of agricultural work. Although most of classifiers required between 30 min and 1 h to extract images for one type of work, one classifier took 2 h.

Two classifiers achieved a recall of 100% for pruning. However, in all other cases, the recall was less than 100%. In this experiment, the classifiers were asked to identify three

types of work in order of pruning, pest insect control, and weed management. The amount of time required to do these tasks varied due to the classifiers' fatigue and familiarity with the task, resulting in less time being required to classify images of insect pest control and weed management.

The recall of the proposed method exceeded 90% and tended to be higher than the recall for manual classification. In contrast, the precision was on the order of 30 to 40% and tended to be lower than the precision for manual classification. The latter result is likely due to the assignment of multiple work types as metadata using the determination method based on time of year ((5) in Fig. 2).

TABLE V. RESULTS OF THE EVALUATION EXPERIMENT (PRUNING)

Pruning							
Classifier	Time required (min)	Total no. of pruning images (A)	No. of images classified (B)	No. of correctly classified images ($A \cap B$)	Precision (%) ($A \cap B / B$)	Recall (%) ($A \cap B / A$)	F-value
1	75	566	995	566	56.9	100	0.73
2	49	566	2250	553	24.6	97.7	0.39
3	70	566	1742	566	32.5	100	0.49
4	48	566	1660	551	33.2	97.3	0.5
5	60	566	206	87	42.2	15.4	0.23
6	120	566	1111	410	36.9	72.4	0.49
7	87	566	621	194	31.2	34.3	0.33
8	73	566	1941	531	27.4	93.8	0.42
9	98	566	889	439	49.4	77.6	0.6
10	103	566	2758	528	19.1	93.3	0.32
11	35	566	588	309	52.6	54.6	0.54
Proposed method	0	566	1806	566	31.3	100	0.47

TABLE VI. RESULTS OF THE EVALUATION EXPERIMENT (INSECT PEST CONTROL)

Insect pest control							
Classifier	Time required (min)	Total no. of insect pest control images (A)	No. of images classified (B)	No. of correctly classified images ($A \cap B$)	Precision (%) ($A \cap B / B$)	Recall (%) ($A \cap B / A$)	F-value
1	44	119	153	83	54.2	69.7	0.61
2	42	119	294	85	28.9	71.4	0.41
3	42	119	424	103	24.3	86.6	0.38
4	44	119	444	63	14.2	52.9	0.22
5	30	119	517	25	4.8	21	0.08
6	25	119	297	62	20.9	52.1	0.3
7	2	119	77	15	19.5	12.6	0.15
8	25	119	735	112	15.2	94.1	0.26
9	5	119	324	83	25.6	69.7	0.37
10	21	119	585	90	15.4	75.6	0.26
11	32	119	209	57	27.3	47.9	0.35
Proposed method	0	119	277	111	40.1	93.3	0.56

TABLE VII. RESULTS OF THE EVALUATION EXPERIMENT (WEED MANAGEMENT)

Weed management							
Classifier	Time required (min)	Total no. of weed management (A)	No. of images classified (B)	No. of correctly classified images ($A \cap B$)	Precision (%) ($A \cap B / B$)	Recall (%) ($A \cap B / A$)	F-value
1	41	78	106	68	64.2	87.2	0.74
2	43	78	193	65	33.7	83.3	0.48
3	26	78	128	64	50	82.1	0.62
4	33	78	648	43	6.6	55.1	0.12
5	60	78	66	64	97	82.1	0.89
6	12	78	35	19	54.3	24.4	0.34
7	30	78	238	17	7.1	21.8	0.11
8	28	78	638	23	3.6	29.5	0.06
9	76	78	842	50	5.9	64.1	0.11
10	31	78	712	46	6.5	59	0.12
11	30	78	81	41	50.6	52.6	0.52
Proposed method	0	78	164	72	43.9	92.3	0.6

In addition, because there is a tradeoff between the recall and the precision, an F-value was calculated and compared to those of the manual classification performed by the 11 classifiers. The F-value for the proposed method was fourth highest for weed management, second highest for insect pest control, and seventh highest for pruning. In the case of pruning, the desired recall of 100% was achieved. Classification of weed management and insect pest control relies on the aforementioned template matching. Some of the images were not clear due dust particles or rain drops on the camera lens. Accordingly, the agricultural chemical sprayer and riding lawnmower could not be identified in all cases, resulting in a recall less than 100%. Even for case of manual classification, the recall was less than 100% because some of the unclear images could not be classified visually even by a human classifier. Except for classifier 5, the recall for weed management was higher for the proposed method than for manual classification.

VI. CONCLUSIONS

The objective of this study was to develop a system for automatically classifying images of agricultural work and animals captured by infrared motion sensor network cameras installed in the field. We proposed a method that employs both image analysis technology and information on the date and time of image capture. We also developed a method for automatically classifying the type of agricultural work. We evaluated the accuracy of the proposed method by comparing the results of automated classification with the results of manual classification. The recall of the proposed method

exceeded 90% for all three types of agricultural work tested (weed management, insect pest control, and pruning), which was equal to or greater than the classification accuracy achieved with manual classification.

Going forward, we plan to investigate methods for determining animal type and, as indicated in Section 2 and Fig. 1, to develop a support system for creating training materials and an adaptive animal deterrence system. In the future, we will explore how the proposed method can be used to support career education and new farmers and other countermeasures to the aging farming population and shortage of new farmers.

ACKNOWLEDGMENT

This work was supported by the Telecommunications Advancement Foundation in Japan.

REFERENCES

- [1] Masanori Takagi, Shohei Yoshida, Takemichi Nakamura, Keizo Yamada, Jun Sasaki, Study on Automatic Recording Method of Agricultural Work for Learning Materials in Elementary School, IPSJ, SSS2012, pp.233-240(2012)
- [2] Ministry of Agriculture, Forestry and Fisheries <http://www.maff.go.jp/j/tokei/sihyo/data/08.html> (confirmed on Feb. 15, 2019)
- [3] Midori box [green box] <https://info.midori-cloud.net/> (confirmed on Feb. 15, 2019)
- [4] Masataka Imura, labeling classes from <http://imura-lab.org/products/labeling/> (confirmed on Feb. 5, 2018)